Descriptive sensory analysis of broiler breast fillets marinated in phosphate, salt, and acid solutions.

Sensory attributes of fully aged broiler breast fillets marinated in a 6% NaCl solution containing 2% sodium tripolyphosphate (2P), 2% citric acid (2C), 2% acetic acid (2A), 1% citric acid plus 1% phosphate solution (1C), or 1% acetic acid solution plus 1% phosphate (1A) were studied. A 6% NaCl (6S) solution with no additives was used as control. Oven-cooked samples (177C degrees oven; 75 degrees C internal temperature) were evaluated by a 9-member trained descriptive analysis sensory panel that rated the intensities of 26 different flavor and texture attributes using 15-point line scales. Data were analyzed using general linear model SAS procedures to determine significant differences (P < or = 0.05) in individual sensory attributes due to marinade treatment. All sensory attributes were scored in the low intensity range (1.5 to 5.0). Brothy, vinegar, and residual particles were the only individual attributes rated significantly different (P < or = 0.05) due to treatment. Multivariate analyses indicated that all sensory attributes formed 2 dimensions that explained 57% of variation in the data. The low intensity values for texture attributes indicated possible negative consequences due to phosphates, salt, and acids when used with fully aged fillets.